Exercises
Probabilistic Artificial Intelligence
Fall 2026

Institute for Machine Learning
Dept. of Computer Science, ETH Zürich
Prof. Dr. Andreas Krause

For questions, please refer to Moodle.
Released on **09/10/2026**

Homework 2 (Gaussian Processes)

General Instructions

- Submission of solutions is not mandatory but solving the exercises are highly recommended. The master solution will be released after the exercise deadline.
- Part of the exercises are available on Moodle as a quiz. These problems are marked with [☑].

Exercise 1: Gaussian Process Kernels

(a) [☑] Given a kernel function $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, which of the following statements about the kernel function are correct?

- ◯ $k(x, x') = (x - x')^2$ is a valid kernel, and it is both stationary and isotropic
- ◯ $k(x, x') = e^{-\frac{|x-x'|}{2}} \cos(x - x')$ is a valid kernel, and it is stationary but not isotropic
- ◯ $k(x, x') = c\left(e^{-|x|}e^{-|x'|}\right)^3 + e^{-|x|}e^{-|x'|} + 1$, with $c \geq 0$ is a valid kernel, and it is not stationary nor isotropic
- ◯ $k(x, x') = xx' + \sin(x - x')$ is a valid kernel but it is not stationary nor isotropic

Let $GP(\mu, k)$ be a Gaussian process indexed by $\mathbb{R}$, which is the set of all real numbers, with mean function $\mu : \mathbb{R} \to \mathbb{R}$ and kernel function $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$. In the following, we assume that we have the following cosine kernel function $k(x, x') = \cos(w(x - x'))$, and zero mean, i.e. $\mu(x) = 0$. We want to model an unknown function $f : \mathcal{X} \to \mathbb{R}$.

(b) [☑] Given three data points $x_1 = 0, x_2 = \frac{\pi}{2w}, x_3 = \frac{\pi}{w}$ for some $w > 0$, we have the marginal distribution $(f(x_1), f(x_2), f(x_3)) \sim \mathcal{N}(0, K)$.
We are now given a noise-free observations for $f(x_3) = 2$.
What is the mean $\mu^p_{1,2}$ and variance $\Sigma^p_{1,2}$ of the posterior distribution

$$(f(x_1), f(x_2))\,|f(x_3) \sim N(\mu^p_{1,2}, \Sigma^p_{(1,2)\times(1,2)})$$

- ◯ $\mu^p_{1,2} = [-2, 0]^T, \Sigma^p_{(1,2)\times(1,2)} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$
- ◯ $\mu^p_{1,2} = [0, -1]^T, \Sigma^p_{(1,2)\times(1,2)} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
- ◯ $\mu^p_{1,2} = [-2, 0]^T, \Sigma^p_{(1,2)\times(1,2)} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
- ◯ $\mu^p_{1,2} = [0, -1]^T, \Sigma^p_{(1,2)\times(1,2)} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

(c) [☑] We now investigate the influence of kernel parameters on the prior distribution of functions. Imagine drawing sample functions from a Gaussian process with mean 0 and an RBF kernel with four pairs of variance and length scale parameters.

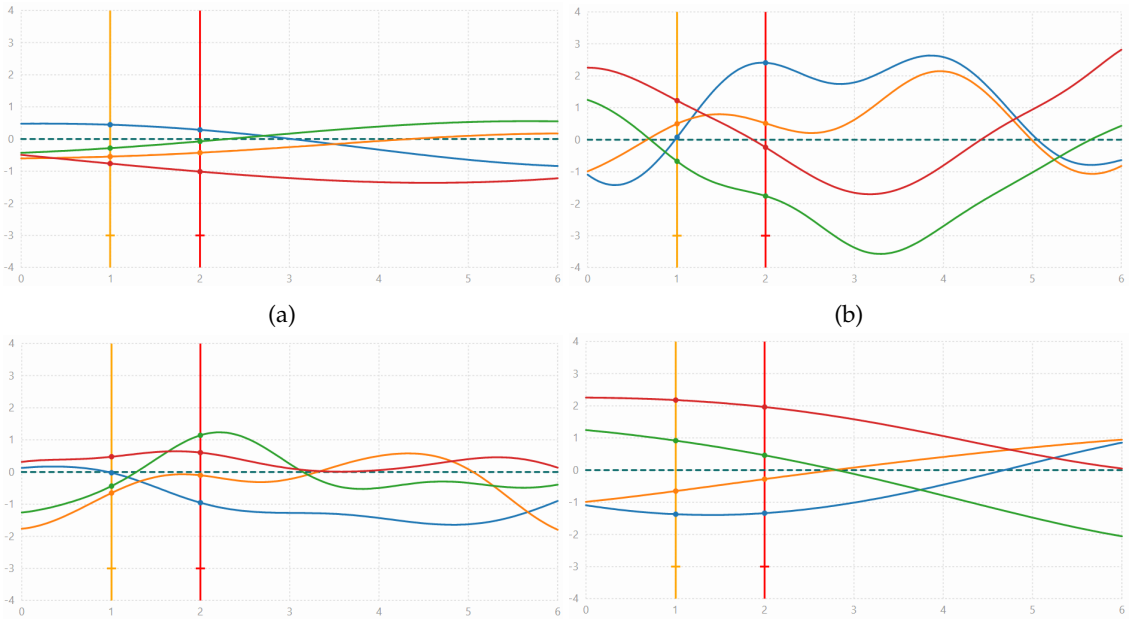
$$k(t, t') = \sigma^2 \exp\left(-\frac{(t - t')^2}{2l^2}\right)$$

1. $\sigma^2 = 0.5, l = 1$.

2. $\sigma^2 = 0.5, l = 4$.
3. $\sigma^2 = 2, l = 1$.
4. $\sigma^2 = 2, l = 4$.

Samples are plotted in Figure 1 in no particular order, each plot for samples drawn from **one** particular pair of parameters.

Applying the results from the previous question, which plot **most likely** corresponds to which pair of parameters?



(a) (b) (c) (d)

Figure 1: Samples drawn from a Gaussian process with mean 0 and RBF kernel using four pairs of parameters. The x-axis represents time t and the y-axis represents X_t. Each plot is generated using a specific pair of variance (σ^2) and length scale (l) parameters. Different colors represent distinct sample functions drawn from the Gaussian process. The function values at $t = 1$ (i.e., X_1) and $t = 2$ (i.e., X_2) are plotted as colored dots for easier comparison.

1. $\sigma^2 = 0.5, l = 1$ ◯ Fig.1a ◯ Fig.1b ◯ Fig.1c ◯ Fig.1d
2. $\sigma^2 = 0.5, l = 4$ ◯ Fig.1a ◯ Fig.1b ◯ Fig.1c ◯ Fig.1d
3. $\sigma^2 = 2, l = 1$ ◯ Fig.1a ◯ Fig.1b ◯ Fig.1c ◯ Fig.1d
4. $\sigma^2 = 2, l = 4$ ◯ Fig.1a ◯ Fig.1b ◯ Fig.1c ◯ Fig.1d

Exercise 2: Gaussian Processes Regression With Linear Kernel

[☑] A *Gaussian process (GP)*, denoted as $GP(\mu, k)$, is a stochastic process specified by some mean function $\mu : \mathcal{X} \to \mathbb{R}$ and some kernel function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$. In this exercise, you will show that Bayesian linear regression yields the same predictions as Gaussian process regression with the linear kernel $k(x, x') = \lambda x^T x'$.

Consider an unknown function $f : \mathcal{X} \to \mathbb{R}$ and a dataset $A = \{(x_1, y_1), \ldots, (x_m, y_m)\}$ consisting of noise-perturbed evaluations $y_i = f(x_i) + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$. Now, our task is to predict the distribution of f at a new point $x^* \in \mathcal{X}$.

GP regression: A priori, we assume $f \sim GP(0, k)$ with linear kernel $k(x, x') = \lambda x^T x'$ (since we want to emulate BLR). In this case, the posterior update for $\mu'(x)$ and $k'(x, x')$ based on the evaluations $y_A = [y_1, \ldots, y_m]^T$ can be computed as follows:

$$\mu'(x) = \mu(x) + k_{x,A}^T (K_{A,A} + \sigma_n^2 I)^{-1} y_A,$$
$$k'(x, x') = k(x, x') - k_{x,A}^T (K_{A,A} + \sigma_n^2 I)^{-1} k_{x',A},$$

where $K_{A,A} \in \mathbb{R}^{m \times m}$ is a matrix with elements $[K_{A,A}]_{i,j} = k(x_i, x_j)$, and $k_{x,A} \in \mathbb{R}^m$ is a vector with elements $[k_{x,A}]_i = k(x, x_i)$.

BLR: In Bayesian linear regression, we assume the linear model $f(x_i) = x_i^T w$ with a prior over the weights $p(w) = \mathcal{N}(0, \sigma_p^2 I)$. In class, we showed that for the evaluations y_A (or $y_{1:n}$ in the lecture notation):

$$p(w \mid y_A) = \mathcal{N}(\bar{\mu}, \bar{\Sigma}),$$
$$\bar{\mu} = \frac{1}{\sigma_n^2} \bar{\Sigma} X^T y_A,$$
$$\bar{\Sigma} = \left(\frac{1}{\sigma_n^2} X^T X + \frac{1}{\sigma_p^2} I \right)^{-1},$$

where $X \in \mathbb{R}^{m \times d}$ consists of the rows $x_1^T, x_2^T, \ldots, x_m^T$.

Given a new point x^*, find the predictive distribution for BLR and GP regression. For which value λ do they coincide?

Hint: One of the Woodbury identities might be useful.

◯ $\lambda = \frac{1}{\sigma_n^2} + \frac{1}{\sigma_p^2}$ ◯ $\lambda = \sigma_p^2$ ◯ $\lambda = \sigma_n^2$ ◯ $\lambda = \frac{1}{\sigma_p^2}$

Exercise 3: Kalman filters

Consider the following dynamical system describing a moving particle:

$$X_{t+1} = X_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_x^2),$$

where $X_t \in \mathbb{R}$ denotes the position of the particle at time $t > 0$, and ε_t is a time-independent and identically distributed (i.i.d.) Gaussian noise term. We would like to track the position of the particle over time; however, this is not straightforward. We cannot directly observe X_t but only measure Y_t, which is perturbed by i.i.d. Gaussian noise η_t:

$$Y_t = X_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma_y^2).$$

To study this process, we divide the problem into prediction and conditioning.

(a) [☑]*Prediction*: Assume that the posterior distribution of X_t is Gaussian with mean μ_t and variance σ_t^2, after having observed $y_{1:t} := \{Y_1 = y_1, \ldots, Y_t = y_t\}$:

$$p(X_t \mid y_{1:t}) = \mathcal{N}(\mu_t, \sigma_t^2).$$

What is the predictive distribution for the particle's position at the next time step, $p(X_{t+1} \mid y_{1:t})$?

○ $\mathcal{N}(0, \sigma_t^2 + t\sigma_y^2)$ ○ $\mathcal{N}(\mu_t, \sigma_y^2 + \sigma_x^2)$ ○ $\mathcal{N}(\mu_t, \sigma_t^2 + \sigma_x^2)$ ○ $\mathcal{N}(\mu_t, \sigma_t^2 + \sigma_y^2)$

(b) [☑]*Conditioning*: once we have the prediction distribution $p(X_{t+1} \mid y_{1:t})$, we would like to understand how acquiring y_{t+1} would help. Here we define the Kalman gain at time t as k_t. For what value k_{t+1} can we write $p(X_{t+1} \mid y_{1:t}, y_{t+1})$ as follows:

$$\begin{aligned} p(X_{t+1} \mid y_{1:t}, y_{t+1}) &= \mathcal{N}(\mu_{t+1}, \sigma_{t+1}^2), \\ \text{with} \quad \mu_{t+1} &= \mu_t + k_{t+1}(y_{t+1} - \mu_t) \\ \text{and} \quad \sigma_{t+1}^2 &= (1 - k_{t+1})(\sigma_x^2 + \sigma_t^2) \end{aligned}$$

○ $k_{t+1} = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_y^2}$ ○ $k_{t+1} = \frac{\sigma_x^2 + \sigma_t^2}{\sigma_x^2 + \sigma_t^2 + \sigma_y^2}$ ○ $k_{t+1} = \frac{\sigma_x^2 + \sigma_t^2 + \sigma_y^2}{\sigma_x^2 + \sigma_t^2}$ ○ $k_{t+1} = \frac{\sigma_x^2 + \sigma_t^2}{\sigma_y^2}$

We now would like to discuss the importance of the Kalman gain k_t. This value plays an important role in the interpretation of the conditioning step.

(c) How do the different parameters σ_x, σ_y, and σ_t influence the value of the Kalman gain?

(d) How does the value of the Kalman gain k_{t+1} give more importance to y_{t+1} or, give more importance to the previous knowledge μ_t? Can you guess why it's called "gain"?

(e) [☑] Now we will show that the Kalman filter can be seen as a Gaussian Process (GP). To this end, we define:

$$f : \mathbb{Z}_+ \to \mathbb{R} \quad \text{such that } f(t) = X_t.$$

Assuming that $X_0 \sim \mathcal{N}(0, \sigma_0^2)$ and $X_{t+1} = X_t + \varepsilon_t$, with $\varepsilon_t \sim \mathcal{N}(0, \sigma_x^2)$, we can show that $f \sim \mathcal{GP}(0, k_{KF})$ with the kernel function k_{KF}:

○ $k_{KF}(t, t') = \sigma_x^2 + \sigma_0^2 \max\{t, t'\}$ ○ $k_{KF}(t, t') = \sigma_x^2 + \sigma_0^2 \min\{t, t'\}$
○ $k_{KF}(t, t') = \sigma_0^2 + \sigma_x^2 \min\{t, t'\}$ ○ $k_{KF}(t, t') = \sigma_0^2 + \sigma_x^2 \max\{t, t'\}$

Exercise 4: Hyperparameters Selection and Marginal Likelihood

Consider an unknown function $f : \mathcal{X} \to \mathbb{R}$ and a dataset $A = \{X, \mathbf{y}\} = \{(x_1, y_1), \ldots, (x_m, y_m)\}$ of noise-perturbed evaluations $y_i = f(x_i) + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$, we make the hypothesis that $f \sim GP(0, k_\theta)$, with zero mean function and covariance function $k_\theta : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$. We are interested in selecting hyperparameters θ by maximizing the marginal likelihood $p(\mathbf{y} \mid X, \theta)$.

(a) [☑] We define $\mathbf{K}_{\mathbf{y},\theta} = \mathbf{K}_{f,\theta} + \sigma_n^2 I$ as the covariance matrix of $\mathbf{y}$ for covariance function k_θ. We also define $\alpha = \mathbf{K}_{\mathbf{y},\theta}^{-1}\mathbf{y}$. What is the value of the marginal likelihood *gradient* with respect to θ_j, i.e. $\frac{\partial}{\partial \theta_j} \log p(\mathbf{y} \mid X, \theta)$:

○ $\frac{1}{2} \operatorname{tr}\left(\left(\alpha\alpha^T - \mathbf{K}_{\mathbf{f},\theta}^{-1}\right) \frac{\partial \mathbf{K}_{\mathbf{f},\theta}}{\partial \theta_j}\right)$ ○ $\frac{1}{2} \operatorname{tr}\left(\left(\alpha\alpha^T - \mathbf{K}_{\mathbf{y},\theta}^{-1}\right) \frac{\partial \mathbf{K}_{\mathbf{y},\theta}}{\partial \theta_j}\right)$
○ $2 \operatorname{tr}\left(\left(\alpha\alpha^T - \mathbf{K}_{\mathbf{y},\theta}^{-1}\right) \frac{\partial \mathbf{K}_{\mathbf{y},\theta}}{\partial \theta_j}\right)$ ○ $2 \operatorname{tr}\left(\left(\alpha\alpha^T - \mathbf{K}_{\mathbf{f},\theta}^{-1}\right) \frac{\partial \mathbf{K}_{\mathbf{y},\theta}}{\partial \theta_j}\right)$

Hint: You can use the following identities in your derivation:

- For any invertible matrix M, you have:

$$\frac{\partial}{\partial \theta_j} M^{-1} = -M^{-1} \frac{\partial M}{\partial \theta_j} M^{-1}$$

- For any symmetric positive definite matrix S, you have:

$$\frac{\partial}{\partial \theta_j} \log |S| = \operatorname{tr}\left(S^{-1} \frac{\partial S}{\partial \theta_j}\right)$$

(b) [☑] We now assume that the covariance function for the noisy targets(i.e. including noise contribution) can be written as $k_{\mathbf{y}}(x, x') = \theta_0 \tilde{k}(x, x')$ with $\tilde{k}$ being a valid kernel independent of θ_0. What is the closed-form solution θ_0^* to the equation $\frac{\partial}{\partial \theta_0} \log p(\mathbf{y} \mid X, \theta) = 0$:

◯ $\frac{1}{m}\mathbf{y}^\top \mathbf{K}_{\mathbf{y}}^{-1}\mathbf{y}$ ◯ $\frac{1}{m}\operatorname{tr}(\tilde{\mathbf{K}}^{-1})$ ◯ $\frac{1}{m}\mathbf{y}^\top \tilde{\mathbf{K}}^{-1}\mathbf{y}$ ◯ $\mathbf{y}^\top \mathbf{K}_{\mathbf{y}}^{-1}\mathbf{y}$

(c) [☑]In the noise-free case, given the covariance function defined as $k_{\theta_0}(x, x') = \theta_0 \hat{k}(x, x')$ with $\hat{k}$ a valid kernel independent of θ_0 and $\hat{k}(x, x') = c\tilde{k}(x, x')$. Let θ_0^* be the optimal parameter with kernel $k_{\theta_0}(x, x') = \theta_0 \tilde{k}(x, x')$. How does optimal parameter θ_0^* scales if we scale labels $\mathbf{y}$ by a scalar s and use the kernel $k_{\theta_0}(x, x') = \theta_0 \hat{k}(x, x')$?

◯ s^2/c
◯ θ_0^* stays identical because it is independent of s and c
◯ sc
◯ It depends of the choice of kernel function $\tilde{k}$

Exercise 5: Sparse Gaussian Processes with Inducing Points and Variational Inference

Consider an unknown function $f : \mathcal{X} \to \mathbb{R}$ with prior $f \sim GP(0, k)$ and a dataset $\{(x_i, y_i)\}_{i=1}^n$ of noise-perturbed evaluations $y_i = f(x_i) + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$. Define

$$A = \{x_1, \ldots, x_n\} \quad , \quad \mathbf{f}_A = [f(x_1), \ldots, f(x_n)]^\top \quad , \quad \mathbf{y} = [y_1, \ldots, y_n]^\top \quad , \quad \mathbf{K}_{AA} = [k(x_i, x_j)]_{i,j}$$

The exact marginal likelihood is given as:

$$p(\mathbf{y}) = \mathcal{N}(\mathbf{y} \mid \mathbf{0}, \mathbf{K}_{AA} + \sigma_n^2 \mathbf{I})$$

Training a GP involves inverting the kernel matrix $(\mathbf{K}_{AA} + \sigma_n^2 \mathbf{I})^{-1}$. This computation costs $\mathcal{O}(n^3)$ time and requires $\mathcal{O}(n^2)$ memory. This makes exact GP regression infeasible beyond tens of thousands of data points. Many approaches to avoid this bottleneck have been proposed. The simplest approximation keeps only $m \ll n$ data points and performs exact GP regression on them. While this reduces the cost to $\mathcal{O}(m^3)$, it ignores $n - m$ observations entirely, which shifts the problem to choosing good points to keep (there are $\binom{n}{m}$ options).

Inducing Points

Better approximations keep all n observations but capture their dependencies through only $m \ll n$ *inducing variables* $\mathbf{u} = [f(\bar{x}_1), \ldots, f(\bar{x}_m)]^\top$: the values of f at *inducing points* $\mathcal{U} = \{\bar{x}_1, \ldots, \bar{x}_m\} \subseteq \mathcal{X}$. The inducing points are not necessarily training inputs and will be learned via optimization. We write

$$\mathbf{K}_{\mathcal{U}\mathcal{U}} = [k(\bar{x}_j \bar{x}_{j'})]_{j,j'} \quad , \quad \mathbf{K}_{A\mathcal{U}} = [k(x_i, \bar{x}_j)]_{i,j} = \mathbf{K}_{\mathcal{U}A}^\top$$
$$\mathbf{k}_{\mathcal{U}}(x) = [k(\bar{x}_1, x), \ldots, k(\bar{x}_m, x)]^\top \quad , \quad \mathbf{Q}_{AA} = \mathbf{K}_{A\mathcal{U}} \mathbf{K}_{\mathcal{U}\mathcal{U}}^{-1} \mathbf{K}_{\mathcal{U}A}$$

(a) The inducing variables are values of the same GP as $\mathbf{f}_A$. Write down the joint prior distribution $p(\mathbf{f}_A, \mathbf{u})$ and derive the conditional distribution $p(\mathbf{f}_A \mid \mathbf{u})$. Then verify that $\int p(\mathbf{f}_A \mid \mathbf{u})\, p(\mathbf{u})\, d\mathbf{u} = p(\mathbf{f}_A)$.

Hint: You can use the following identities:

- If $\begin{bmatrix}\mathbf{a}\\ \mathbf{b}\end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix}A & C\\ C^\top & B\end{bmatrix}\right)$, then $\mathbf{a} \mid \mathbf{b} \sim \mathcal{N}\big(CB^{-1}\mathbf{b},\ A - CB^{-1}C^\top\big)$
- If $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, P)$ and $\mathbf{v} \mid \mathbf{u} \sim \mathcal{N}(M\mathbf{u}, R)$, then $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, MPM^\top + R)$

Part (a) shows that introducing $\mathbf{u}$ is exact: so far, nothing has been approximated. As shown by Quiñonero-Candela and Rasmussen (2005), essentially all sparse GP methods differ only in how they approximate the conditional $p(\mathbf{f}_A \mid \mathbf{u})$. The *fully independent training conditional* (FITC) approximation, proposed by Snelson and Ghahramani (2006), keeps every marginal conditional $p(f_i \mid \mathbf{u})$ exact but assumes that the f_i are conditionally independent given $\mathbf{u}$:

$$q(\mathbf{f}_A \mid \mathbf{u}) = \prod_{i=1}^{n} p(f_i \mid \mathbf{u}) = \mathcal{N}\big(\mathbf{K}_{A\mathcal{U}}\mathbf{K}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{u},\ \Lambda_f\big), \qquad \Lambda_f = \operatorname{diag}(\mathbf{K}_{AA} - \mathbf{Q}_{AA}).$$

Replacing $p(\mathbf{f}_A \mid \mathbf{u})$ by $q(\mathbf{f}_A \mid \mathbf{u})$ changes the marginal likelihood, which becomes the training objective of FITC.

(b) Compute $q(\mathbf{y} \mid \mathbf{u}) = \int p(\mathbf{y} \mid \mathbf{f}_A)\, q(\mathbf{f}_A \mid \mathbf{u})\, d\mathbf{f}_A$ and show that it factorizes over the data points. Then show that the approximate marginal likelihood is

$$q(\mathbf{y}) = \int q(\mathbf{y} \mid \mathbf{u})\, p(\mathbf{u})\, d\mathbf{u} = \mathcal{N}\big(\mathbf{y} \mid \mathbf{0},\ \mathbf{Q}_{AA} + \Lambda\big), \qquad \Lambda := \Lambda_f + \sigma_n^2\mathbf{I}.$$

Finally, show that the approximation is exact for $m = n$ and $\mathcal{U} = A$, i.e. $q(\mathbf{y}) = p(\mathbf{y})$.

Hint: Given $\mathbf{f}_A$, write $\mathbf{y} = \mathbf{f}_A + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2\mathbf{I})$. Also remember the hints from (a).

The result has a simple interpretation. The diagonal of $\mathbf{Q}_{AA} + \Lambda_f$ equals that of $\mathbf{K}_{AA}$, so FITC keeps the prior variance of every $f(x_i)$ exact and only replaces the covariances between different points by those of the rank-m matrix $\mathbf{Q}_{AA}$. Dropping the diagonal correction altogether, i.e. setting $q(\mathbf{f}_A \mid \mathbf{u}) = \mathcal{N}(\mathbf{K}_{A\mathcal{U}}\mathbf{K}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{u}, 0)$, gives the older *subset of regressors* approximation $q(\mathbf{y}) = \mathcal{N}(\mathbf{0}, \mathbf{Q}_{AA} + \sigma_n^2\mathbf{I})$. This is exact GP regression with the Nyström kernel $\tilde{k}(x, x') = \mathbf{k}_{\mathcal{U}}(x)^\top \mathbf{K}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{k}_{\mathcal{U}}(x')$.

In Exercise 4, the hyperparameters θ were selected by maximizing the marginal likelihood. FITC treats the inducing points in exactly the same way: $\mathcal{U}$ and θ (kernel parameters and σ_n^2) are found jointly by maximizing $\log q(\mathbf{y} \mid \mathcal{U}, \theta)$ with a gradient-based optimizer. Compared to the primitive subset-of-data approach, this replaces the discrete choice of m data points with a continuous optimization over $\mathcal{U}$. The derivation in Exercise 4(a) only used that $\mathbf{y} \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_\mathbf{y})$ with a symmetric positive definite covariance that depends differentiably on the parameter. We can therefore use the formula derived above as is, with $\mathbf{K}_\mathbf{y} = \mathbf{Q}_{AA} + \Lambda$. For any scalar parameter η (an entry of some $\bar{x}_j$ or of θ),

$$\frac{\partial}{\partial \eta} \log q(\mathbf{y}) = \frac{1}{2} \operatorname{tr}\left(\left(\alpha\alpha^\top - \mathbf{K}_\mathbf{y}^{-1}\right) \frac{\partial \mathbf{K}_\mathbf{y}}{\partial \eta} \right), \qquad \alpha = \mathbf{K}_\mathbf{y}^{-1}\mathbf{y}.$$

In the following, let $\mathcal{X} = \mathbb{R}^d$.

(c) Let $\eta = (\bar{x}_j)_r$ be the r-th coordinate of the j-th inducing input. Express $\partial\mathbf{K}_\mathbf{y}/\partial\eta$ in terms of $\partial\mathbf{K}_{A\mathcal{U}}/\partial\eta$ and $\partial\mathbf{K}_{\mathcal{U}\mathcal{U}}/\partial\eta$, and state which entries of these two matrices can be non-zero.

Hint: For any invertible matrix M, $\frac{\partial}{\partial\eta}M^{-1} = -M^{-1}\frac{\partial M}{\partial\eta}M^{-1}$.

Variational Learning of Inducing Points

This part of the exercise requires knowledge of variational inference. Revisit it once variational inference has been introduced in the lecture.

The FITC objective is the marginal likelihood of a *modified* model. Because the GP prior was replaced by the approximation q, the inducing points effectively act as additional kernel hyperparameters, and nothing prevents $\log q(\mathbf{y})$ from exceeding the exact $\log p(\mathbf{y})$ at any value of $\mathbf{y}$. With many free parameters, this can lead to overfitting. Titsias (2009) instead keeps the exact GP model and approximates its *posterior*, which can be written as

$$p(\mathbf{f}_A, \mathbf{u} \mid \mathbf{y}) = p(\mathbf{f}_A \mid \mathbf{u}, \mathbf{y})\, p(\mathbf{u} \mid \mathbf{y}).$$

Titsias approximates both factors. He replaces the conditional $p(\mathbf{f}_A \mid \mathbf{u}, \mathbf{y})$ by the prior conditional $p(\mathbf{f}_A \mid \mathbf{u})$, using the assumption that given the inducing variables, the training samples $\mathbf{y}$ do not carry any additional information about $\mathbf{f}_A$ (this step is exact if $\mathbf{K}_{AA} = \mathbf{Q}_{AA}$). He also replaces the computationally infeasible posterior $p(\mathbf{u} \mid \mathbf{y})$ by a free variational distribution $\varphi(\mathbf{u})$. This gives

$$q(\mathbf{f}_A, \mathbf{u}) = p(\mathbf{f}_A \mid \mathbf{u})\, \varphi(\mathbf{u}).$$

Both approximations concern only the posterior; unlike FITC, the underlying GP model is not modified. Optimizing φ analytically leads to the objective

$$\mathcal{F}(\mathcal{U}, \theta) = \log \mathcal{N}\big(\mathbf{y} \mid \mathbf{0},\ \mathbf{Q}_{AA} + \sigma_n^2 \mathbf{I}\big) - \frac{1}{2\sigma_n^2} \operatorname{tr}\big(\mathbf{K}_{AA} - \mathbf{Q}_{AA}\big).$$

It is a lower bound on the exact log marginal likelihood,

$$\mathcal{F}(\mathcal{U}, \theta) \le \log p(\mathbf{y}),$$

for every choice of $\mathcal{U}$. Compared with FITC, the correction $\mathbf{K}_{AA} - \mathbf{Q}_{AA}$ no longer enters the covariance but appears as a penalty.

(d) Fix $\mathbf{u}$ and let $\mathbf{f}_A \sim p(\mathbf{f}_A \mid \mathbf{u})$ from (a). Show that

$$\mathbb{E}_{p(\mathbf{f}_A \mid \mathbf{u})}\big[\log \mathcal{N}(\mathbf{y} \mid \mathbf{f}_A, \sigma_n^2 \mathbf{I})\big] = \log \mathcal{N}\big(\mathbf{y} \mid \mathbf{K}_{A\mathcal{U}} \mathbf{K}_{\mathcal{U}\mathcal{U}}^{-1} \mathbf{u},\ \sigma_n^2 \mathbf{I}\big) - \frac{1}{2\sigma_n^2} \operatorname{tr}\big(\mathbf{K}_{AA} - \mathbf{Q}_{AA}\big).$$

Hint: If a random vector $\mathbf{f}_A$ has mean $\boldsymbol{\mu}$ and covariance S, then $\mathbb{E}\|\mathbf{y} - \mathbf{f}_A\|^2 = \|\mathbf{y} - \boldsymbol{\mu}\|^2 + \operatorname{tr}(S)$.

(e) Show that the penalty $\frac{1}{2\sigma_n^2} \operatorname{tr}(\mathbf{K}_{AA} - \mathbf{Q}_{AA})$ is non-negative and vanishes for $m = n$ and $\mathcal{U} = X$. Conclude that $\mathcal{F}(X, \theta) = \log p(\mathbf{y})$.

(f) Show that for the variational distribution $q(\mathbf{f}_A, \mathbf{u}) = p(\mathbf{f}_A \mid \mathbf{u})\, \varphi(\mathbf{u})$, the ELBO of the model $p(\mathbf{y}, \mathbf{f}_A, \mathbf{u}) = p(\mathbf{y} \mid \mathbf{f}_A)\, p(\mathbf{f}_A \mid \mathbf{u})\, p(\mathbf{u})$ equals

$$\mathcal{L}(\varphi) = \mathbb{E}_{\varphi(\mathbf{u})}\Big[\mathbb{E}_{p(\mathbf{f}_A \mid \mathbf{u})}\big[\log p(\mathbf{y} \mid \mathbf{f}_A)\big]\Big] - \mathrm{KL}\big(\varphi \,\|\, p(\mathbf{u})\big).$$

Then use (e) to show that $\max_\varphi \mathcal{L}(\varphi) = \mathcal{F}(\mathcal{U}, \theta)$.

Hint: For a non-negative function g with $0 < \int g(\mathbf{u})\, d\mathbf{u} < \infty$, the maximum of $\int \varphi(\mathbf{u}) \log \frac{g(\mathbf{u})}{\varphi(\mathbf{u})}\, d\mathbf{u}$ over probability densities φ is $\log \int g(\mathbf{u})\, d\mathbf{u}$.

By (f), $\mathcal{F}$ equals $\log p(\mathbf{y})$ at $\mathcal{U} = X$. Since $\mathcal{F} \le \log p(\mathbf{y})$ for every choice of $\mathcal{U}$, maximizing $\mathcal{F}$ over $\mathcal{U}$ can only bring the variational approximation closer to the exact GP in terms of the variational bound. More precisely, for fixed θ,

$$\log p(\mathbf{y}) - \mathcal{F} = \mathrm{KL}\big(q(\mathbf{f}_A, \mathbf{u}) \,\|\, p(\mathbf{f}_A, \mathbf{u} \mid \mathbf{y})\big)\,,$$

so increasing $\mathcal{F}$ is equivalent to decreasing the KL divergence between $q(\mathbf{f}_A, \mathbf{u})$ and the exact posterior. FITC has neither this lower-bound property nor the corresponding KL-divergence interpretation: although its objective also equals $\log p(\mathbf{y})$ at $\mathcal{U} = X$ (part (e)), for other choices of $\mathcal{U}$ it may exceed $\log p(\mathbf{y})$.